

THE GEOGRAPHY OF GRIEF: MAPPING 20 YEARS OF OVERDOSE DEATHS

Drug overdose death rates tripled equally across Rural, Suburban & Urban America from 2003–2021.
Yet Suburban counties consistently bear the highest burden - and West Virginia is a crisis within a crisis.

AUDIENCE: Public health policymakers & informed public

ANGLE: Suburban counties lead - not urban, not rural. A truth hiding in plain sight.

3x

rise in death rates since 2003
Identical across Rural, Suburban & Urban - no group was spared

279

counties at crisis level (50+/100k)
Concentrated in Appalachia, Mountain West & New England

6 of 8

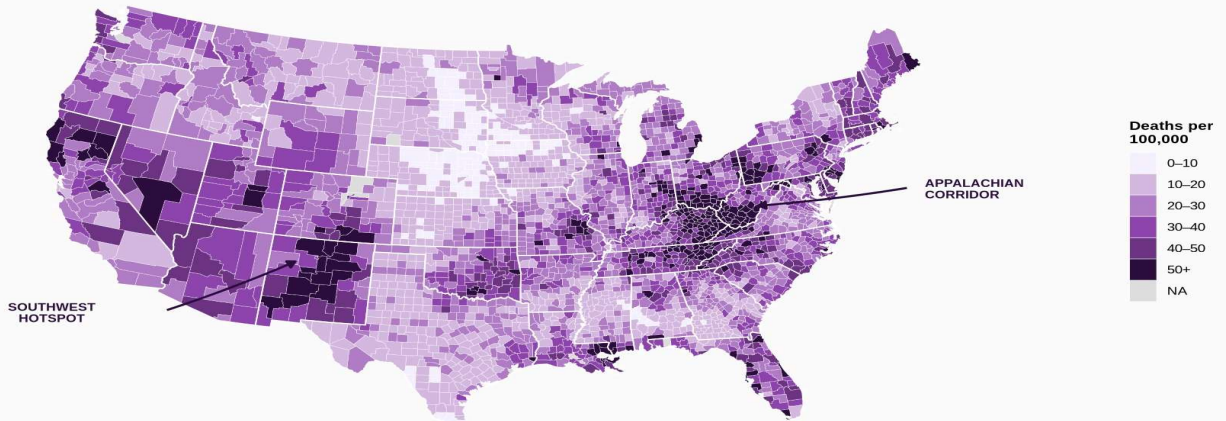
worst counties are in West Virginia
McDowell Co. peaks at 164.6 - more than 5x the national average

#1

New England – highest of all 9 regions
Defies every assumption about where the crisis lives

Where the Crisis Lives: County-Level Overdose Deaths in 2021

279 counties exceed 50 deaths per 100,000 — Appalachia & Southwest hardest hit

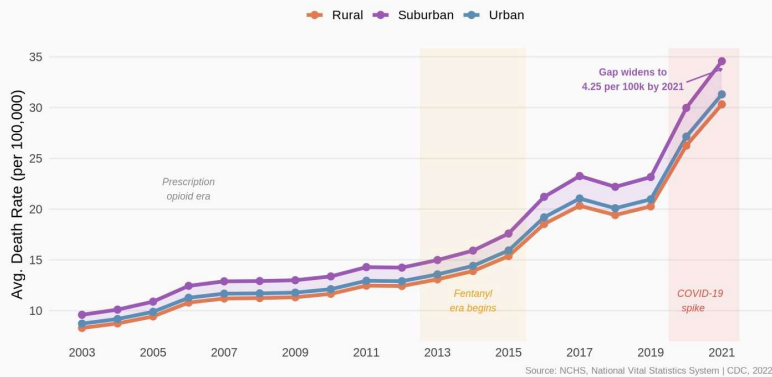


What you're seeing: County-level drug overdose death rates in 2021 across all 3,136 U.S. counties. Deep geographic clustering is visible - the Appalachian corridor (WV, KY, OH, TN) and the Southwest (NM, CO, AZ) show the darkest counties. 279 counties exceed 50 deaths per 100,000 - more than 5x the national average. Gray = insufficient data.

SUPPORTING CHART 1 — TREND OVER TIME

The Overdose Crisis Has Always Had a Suburban Address

Population-weighted avg. drug overdose death rate per 100,000 | Suburban counties consistently highest since 2003, with the gap tripling by 2021

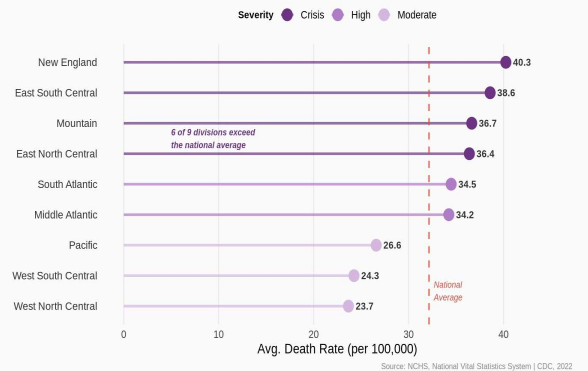


What you're seeing: Suburban counties (purple) have led overdose death rates every year since 2003. The fentanyl era (2013–15) and COVID-19 (2020) accelerated all three groups equally. But the Suburban–Rural gap tripled to 4.25 per 100,000 by 2021.

SUPPORTING CHART 2 — REGIONAL RANKING

New England Tops the Crisis: Not Where You'd Expect

Population-weighted avg. overdose death rate by Census Division, 2021
Above 30 per 100,000 = Crisis level

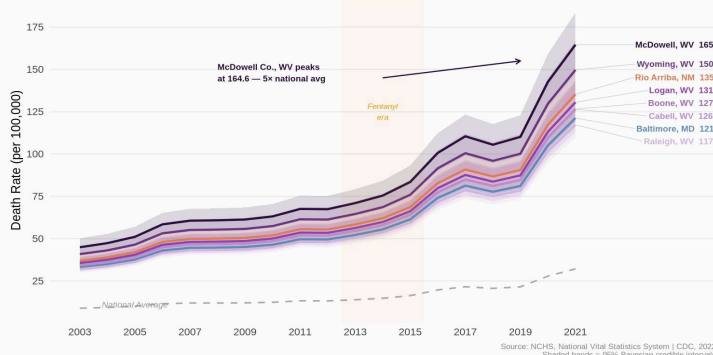


What you're seeing: New England leads all 9 Census Divisions at 40.3 deaths per 100,000 - far exceeding expectations. 6 of 9 divisions exceed the national average (red dashed line), confirming the crisis is truly nationwide, not regionally isolated.

SUPPORTING CHART 3 — SPOTLIGHT: WEST VIRGINIA

West Virginia Is a Crisis Within a Crisis

Drug overdose death rates for the 8 hardest-hit U.S. counties, 2003–2021
6 of 8 are in West Virginia — McDowell County peaks at 5x the national average



What you're seeing:

The 8 hardest-hit U.S. counties by 2021 death rate. 6 of 8 are in West Virginia. McDowell County, WV peaks at 164.6 per 100,000 — more than 5x the national average of ~32. Shaded bands show 95% Bayesian credible intervals. All counties track far above the national average (dashed gray), exposing West Virginia as a state-level humanitarian emergency.

KEY TAKEAWAY FOR POLICYMAKERS

The overdose crisis is not a city problem or a rural problem; it is an everywhere problem that has been growing uniformly since 2003. Federal and state resources must be rebalanced: Suburban counties have led death rates since the very beginning, West Virginia requires emergency-level intervention, and New England's top-ranking defies every geographic assumption. Data-driven reallocation of prevention and treatment funding is urgently needed.

Data: NCHS Drug Poisoning Mortality by County, United States 2003–2021
CDC / National Center for Health Statistics, 2022
Visualization: Module 8 Data Storytelling Final Project